Miraj Chamara Samarakkody

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SUMMARY

Assistant Professor of Mathematics at Tougaloo College with a Ph.D. from Texas Tech University. Research spans pure and applied differential geometry, with active work on variational problems. Principal Investigator on NIH- and NSF-funded research projects, with published work in differential geometry and open-source contributions to mathematical software.

RESEARCH INTERESTS

Differential geometry; minimal surfaces; generalized elastic curves; variational problems; geometric analysis; mathematical physics; applications to biology and data science; formalization of mathematics in Lean 4; Julia package development for machine learning and differential geometry.

EDUCATION

Ph.D. in Mathematics, Fall 2021 – Spring 2025
Texas Tech University, Lubbock, TX
Advisors: Dr. Hung Tran (Chair), Dr. Álvaro Pámpano (Co-chair)

B.Sc. (Honors) Special in Mathematics, 2014 – 2018
University of Peradeniya, Sri Lanka
Advisor: Dr. Shelton Perera

PROFESSIONAL EXPERIENCE

Assistant Professor, Spring 2025 – Present
Department of Mathematics and Computer Science, Tougaloo College, MS

Graduate Part-Time Instructor, Fall 2024
Department of Mathematics and Statistics, Texas Tech University, TX

Research Assistant, Spring 2022 – Summer 2024
Department of Mathematics and Statistics, Texas Tech University, TX
Mentors: Dr. Hung Tran and Dr. Álvaro Pámpano

Graduate Teaching Assistant, Fall 2021
Department of Mathematics and Statistics, Texas Tech University, TX

Lecturer, 2019 – Fall 2021
Department of Mathematics, University of Moratuwa, Sri Lanka

SHORT VISITS

Research Mentor, Virtual Math Circle, Department of Mathematics, Louisiana State University
Summer 2025

TECHNICAL SKILLS

Programming Languages: Python, Julia, C
Mathematical Software: Mathematica, MATLAB

Typesetting: LaTeX
Theorem Proving: Lean 4
Design and Media: Blender, Adobe Illustrator, Adobe Premiere Pro
Productivity: Microsoft Office

TEACHING

Tougaloo College

- MAT103 - College Algebra; Summer 2025, Summer 2026
- MAT221 - Calculus I; Fall 2025, Spring 2026
- MAT222 - Calculus II; Spring 2025, Spring 2026
- MAT316 - Differential Equations; Fall 2025
- MAT326 - Introduction to Probability; Fall 2025
- MAT414 - Modern Algebra; Spring 2025
- MAT426 - Advanced Calculus; Spring 2025, Fall 2025
- MAT429 - Complex Variables; Spring 2026
- MAT434 - Theory of Mathematical Statistics; Spring 2025, Spring 2026

Texas Tech University

- MATH 1452 - Calculus II with Applications; Fall 2024

GRANTS

- **Principal Investigator**, Institutional Development Award (IDeA), National Institute of General Medical Sciences, National Institutes of Health, Grant Number P20GM103476. Amount: $29,317.
  Project: Multivariate Statistical Analysis and Machine Learning Approaches for Heart Disease Prediction and Risk Factor Identification.
- **Recipient**, HBCU UP Implementation Project, Award Number 2510537.
- **Principal Investigator**, SCALE Program (Partnership with Purdue University). Amount: $73,094.77.
  Project: Microelectronic Workforce Development Project for Radiation-Hardening.

FELLOWSHIPS

- **Dr. Shelby Hildebrand Graduate Math Fellowship**, Texas Tech University, 2023 – 2024. Amount: $10,000.
- **AT&T Graduate Fellowship**, Texas Tech University, 2021 – 2025. Amount: $16,000.

AWARDS AND HONORS

- **Create Possible Scholarship**, Texas Tech University, 2022.
- **University Prize for Academic Excellence**, University of Peradeniya, 2018.
- **Peradeniya University Alumni Australia Victoria Chapter Scholarship for Academic Excellence in Mathematics**, 2015.

PUBLICATIONS

- M. Samarakkody. "Formalizing the Classical Isoperimetric Inequality in the Two-Dimensional Case." Submitted.

- Y. Li, E. Güler, M. Toda, M. Samarakkody. “Geometric Properties of Hypersurfaces of Right Conoids in Minkowski Spacetime.” Submitted.
- A. Pámpano, M. Samarakkody, H. Tran. “Closed p-Elastic Curves in Spheres of $\mathbb{L}^3$.” Journal of Mathematical Analysis and Applications, 542(2):129147, 2025.
- S.R.S.M.C. Samarakkody, P.G.R.S. Ranasinghe, A.A.S. Perera. “Geometric Behavior of a Certain Class of Quotients of Finite Blaschke Products.” International Conference on Multidisciplinary Engineering (ICMME), Post Graduate Institute of Science, University of Peradeniya, Vol. 1, 2019.

SOFTWARE

- **ParametricCurves.jl** - Julia package for visualizing and analyzing parametric curves.
  Repository: github.com/mirajcs/ParametricCurves.jl
  DOI: 10.5281/zenodo.20140761

PRESENTATIONS

Conference Talks

- “Closed p-Elastic Curves in 2-Space Forms.” AMS Sectional Meeting (Meeting No. 1198), University of Texas at San Antonio, September 14–15, 2024.
- “Closed p-Elastic Curves in Spheres of Lorentz-Minkowski Space.” Texas Analysis and Mathematical Physics Symposium, Texas A&M University, February 10, 2024.
- “Geometric Behavior of a Certain Class of Quotients of Finite Blaschke Products.” ICMME, Post Graduate Institute of Science, University of Peradeniya, March 2019.

Posters

- “Formalizing the Isoperimetric Inequality in Lean 4.” XX Red Raider Minisymposium - Geometric Analysis and Applications, Texas Tech University, April 25, 2026.
- “Closed p-Elastic Curves in 2-Space Forms.” Texas Geometry and Topology Conference (68th Meeting), Texas A&M University, College Station, November 8–10, 2024.

Seminars

- “From Manifold to Machines: Formalizing Differential Geometry in Lean.” Probability, Differential Geometry and Mathematical Physics Seminar, Texas Tech University, April 24, 2026.
- “On Some Variational Problems in Curves and Surfaces.” Probability, Differential Geometry and Mathematical Physics Seminar, Texas Tech University, January 21, 2025.
- “On Some Variational Problems in Differential Geometry.” Probability, Differential Geometry and Mathematical Physics Seminar, Texas Tech University, November 1, 2023.
- “Minimal Surfaces.” Probability, Differential Geometry and Mathematical Physics Seminar, Texas Tech University, November 2, 2022.

LEADERSHIP AND COMMUNITY INVOLVEMENT

- Member, Commission on Animal Care - Institutional Review Board (IRB), Tougaloo College, 2025 – 2026.
- Judge, Kincheloe Research Symposium, Tougaloo College.
- Counselor, Summer Chess Camp 2024, Texas Tech University.
- Judge, Arts and Humanities Conference 2023, Texas Tech University.

- Mentor, Math Circle 2023 – 2024, Department of Mathematics and Statistics, Texas Tech University.

PROFESSIONAL MEMBERSHIPS

- Member, American Mathematical Society, 2021 – 2025.
- Member, SIAM Texas Tech University Chapter, 2021 – 2025.